SCIENCE

Entropy and the Arrow of Time

Why can you unscramble an egg in a video played backward, but never in real life? The answer — entropy — may be the deepest reason time only flows one way.

Source: entropy-arrow-of-time.md **Date:** 2026-05-27 **Author:** Ankit Gupta

Tags: thermodynamics, entropy, time, physics

Drop a glass and it shatters. You never see the shards leap back together. Yet almost every fundamental law of physics works the same forwards and backwards in time — so why does the world have such an obvious *direction*? The answer is one of the most profound ideas in science: **entropy**.

Entropy is about counting

Entropy is often called “disorder,” but a better word is **possibilities**. It measures how many microscopic arrangements (positions and motions of every particle) correspond to the same macroscopic state.

Consider a deck of cards. There’s exactly **one** arrangement we call “perfectly sorted,” but there are about 8×10^{67} shuffled arrangements. Shuffle a sorted deck and it becomes disordered — not because some force pushes it, but simply because there are overwhelmingly more disordered arrangements than ordered ones. Order is rare; disorder is common. Randomness almost always lands you in the vast sea of “messy.”

The Second Law of Thermodynamics

This is the famous **Second Law**: in an isolated system, entropy tends to increase over time. Not because it’s forbidden to decrease, but because decreasing is astronomically unlikely. A shattered glass reassembling would require every molecule to move *just so* — one arrangement out of an incomprehensible number. The universe doesn’t ban it; it just never happens on the timescales we’d ever observe.

Why this gives time a direction

Here’s the deep part. The microscopic laws of physics are nearly **time-symmetric** — run a movie of two colliding particles backwards and it still obeys the laws. So where does the one-way arrow of everyday life come from?

From entropy. The one thing that reliably distinguishes past from future is that **entropy was lower in the past and is higher in the future**. “Forward in time” is, essentially, “the direction in which entropy increases.” Your memories, aging, the cooling of coffee, the very sense that time *flows* — all point the same way because all are entropy increasing. This is called the **thermodynamic arrow of time.**

The cosmic puzzle

This raises a staggering question: if high entropy is overwhelmingly likely, why was the *early universe* in such an extraordinarily low-entropy state to begin with? That low-entropy beginning is what makes everything since possible — it's the reservoir of order that life, stars, and structure spend down. Why the universe started so orderly remains one of the great open questions in cosmology.

Order isn't impossible — it's local

If entropy always increases, how do living things, cities, and crystals form? Because the Second Law applies to *isolated* systems. You can create local order by exporting disorder elsewhere. Your body maintains exquisite organization by consuming low-entropy food and radiating heat (high entropy) into your surroundings. Life is a beautiful, temporary eddy of order in a universe flowing toward disorder.

The takeaway

Entropy explains why the future looks different from the past, why perpetual motion is impossible, and why the simple act of remembering yesterday but not tomorrow is built into the laws of physics. Time's arrow isn't a separate ingredient of reality — it's statistics, written across the whole cosmos.

Does the low-entropy early universe demand an explanation, or is it just a brute fact? Weigh in below.